

CS4063 - Natural Language Processing

Due Date: Friday, September 6th by 10:00am.

Assignments are to be done individually. No late assignments will be accepted.

Submissions that do not comply with the specifications given in this document will not be marked and a zero grade will be assigned.

You are required to submit a one or two JPEG files. Please do not submit zip files in this assignment. You should be able download the Urdu dataset to be used in later assignments.

Audio Data Curation

1 Introduction

Speech recognition and synthesis is an important area in Natural Language Processing. It involves processing data in various modalities such as voice commands, scripted speech, and conversational speech. Preparing high quality data for downstream tasks is essential for the success of NLP applications. In this assignment, we will learn how data is made useful through high quality labeling. For the accuracy of training NLP models, accurately labeled data is of extreme importance. For increasing accuracy and generalizability of NLP models we need to increase the quality and quantity of labeled data. Your task in this assignment is to contribute to open source dataset being curated by Mozilla Common Voice.

Common Voice is a crowdsourcing project started by Mozilla to create a free database for speech recognition software. The project is supported by volunteers who record sample sentences with a microphone and review recordings of other users.

2 Assignment Task

Currently there is about a couple of hundred hours of Urdu voice data contributed by a small number of volunteers out of which only 64 hours has been validated. The task is to contribute to this Urdu dataset on Common Voice. Urdu is spoken by 70M native speakers and yet Urdu datasets for NLP are scarcely available. In this assignment we will attempt to contribute to the open source CommonVoice dataset. Common Voice is part of Mozilla's initiative to help teach machines how real people speak.

For this assignment, you will be contributing at-least **one hour** of your voice to Common Voice (by Speaking) and validating at-least **an hour** of voice from other speakers.

Additionally, you will also be contributing and validating Urdu text sentences to Common Voice. You will provide evidence that you have added at-least 200 public domain Urdu sentences to CommonVoice and reviewed at-least 200 sentences for correctness according to the guidelines.

3 Guidelines

- Create an account on [Common Voice](#), and a profile. Set 'Leaderboard Visibility' to Visible.
- Familiarize yourself with Common Voice guidelines and policies. Remember that your contributions will be validated by other users as well as your instructors.
- Locate Urdu among launched languages and start contributing!
- **Submission:** You will be submitting a snapshot of your 'Stats' clearly showing your account id, your recordings and your validations.
- **Grading:** This assignment will be graded on a relative scheme with the highest contributors getting the highest marks.

4 Useful Links

- Common Voice: <https://commonvoice.mozilla.org/>
- Sentence Collector Write: <https://commonvoice.mozilla.org/en/write>
- Sentence Collector Review: <https://commonvoice.mozilla.org/en/review>
- Urdu Keyboard: <https://www.lexilogos.com/keyboard/urdu.htm>

Honor Policy

This assignment is a learning opportunity that will be evaluated based on your ability to think in a group setting, work through a problem in a logical manner and write a research report on your own. You may however discuss verbally or via email the assignment with your classmates or the course instructor, but you are to write the actual report for this assignment without copying or plagiarizing the work of others. You may use the Internet to do your research, but the written work should be your own. **Plagiarized reports or code will get a zero.** If in doubt, ask the course instructor.